

Module 4: Route 53 Assignment

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Problem Statement:

You work for XYZ Corporation that uses on premise solutions and some limited number of systems. With the increase in requests in their application, the load also increases. So, to handle the load the corporation has to buy more systems almost on a regular basis. Realizing the need to cut down the expenses on systems, they decided to move their infrastructure to AWS.

Tasks to do:

1. To use the Route 53 hosted zone created in the assignment.
2. To route the traffic to an EC2 instance with an Apache web server running in it using its IP address.

Solution:

Task 0 – Prerequisite

Started the EC2 instance from previous assignment

A. Created EC2 instances

EC2 → Instances → Launch instance

Selected Configuration and launched 1 EC2 server.

ELB-Server-1

EC2 > Instances > Launch an instance

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

ELB-Server-1

Add additional tags

▼ Application and OS Images (Amazon Machine Image)

Info ⓘ

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

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Red Hat

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debian

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Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-0fef201115ee936 (64-bit (x86), uefi-preferred) / ami-0eb45f74aa8a20238 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

EC2 > Instances > i-03f191509e3878db6

Instance summary for i-03f191509e3878db6 (ELB-Server-1)

Info ⓘ

Updated less than a minute ago

Connect

Instance state

Actions

Instance ID

i-03f191509e3878db6

IPv6 address

–

Hostname type

IP name: ip-172-31-11-238.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

3.239.18.96 [Public IP]

IAM role

–

IMDSv2

Required

Operator

Public IPv4 address

3.239.18.96 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-11-238.ec2.internal

Instance type

t3.micro

VPC ID

vpc-030682f2c586ad6b1

Subnet ID

subnet-02791cbf0e8268fb9

Instance ARN

arn:aws:ec2:us-east-1:430776404757:instance/i-03f191509e3878db6

Private IPv4 addresses

172.31.11.238

Public DNS

ec2-3-239-18-96.compute-1.amazonaws.com | open address

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

–

Managed

false

B. Installed Web-Server in the EC2 server

Connected to ELB-Server-1

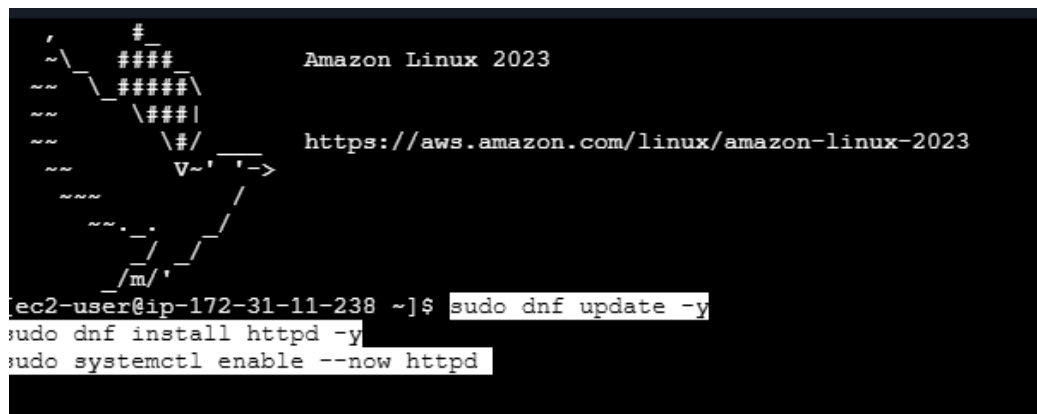
ELB-Server-1 → Connect

EC2 Instance Connect → Connect

Ran the following bash to install Apache webserver and enable it:

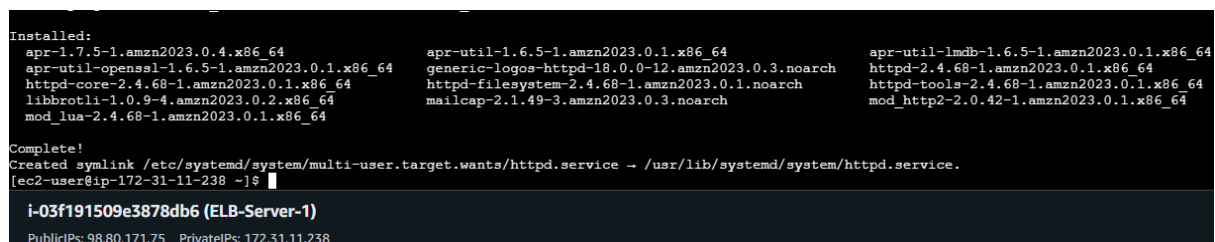
On terminal ran:

```
sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```

A terminal window on an Amazon Linux 2023 instance. The prompt is [ec2-user@ip-172-31-11-238 ~]\$. The user has entered the commands: sudo dnf update -y, sudo dnf install httpd -y, and sudo systemctl enable --now httpd. The output shows the packages being installed and the service being enabled.

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-11-238 ~]$ sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```

A terminal window showing the output of the installation. It lists the installed packages and the symlink created for the httpd service. The prompt is [ec2-user@ip-172-31-11-238 ~]\$.

```
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.5-1.amzn2023.0.1.x86_64      apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch      httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-11-238 ~]$
```

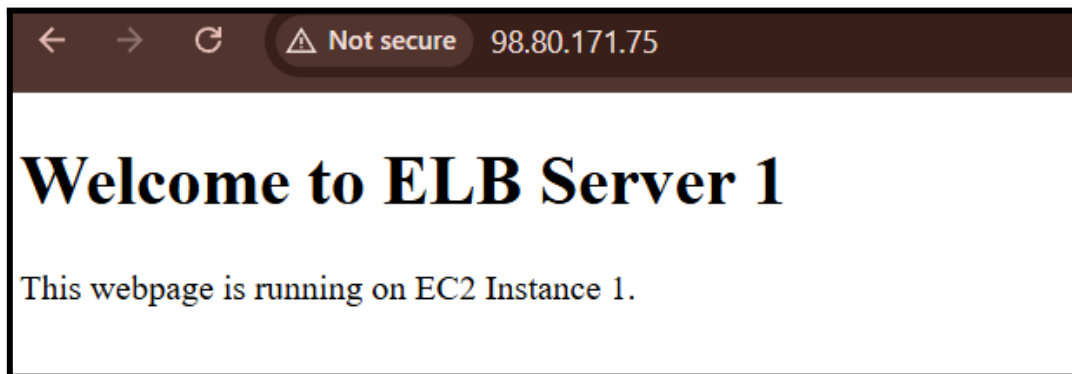
C. Created Server 1 webpage

Bash to open index.html : sudo nano /var/www/html/index.html

edited index.html to:

```
<!DOCTYPE html>
<html>
<head>
  <title>ELB Server 1</title>
</head>
<body>
  <h1>Welcome to ELB Server 1</h1>
  <p>This webpage is running on EC2 Instance 1.</p>
</body>
</html>
```

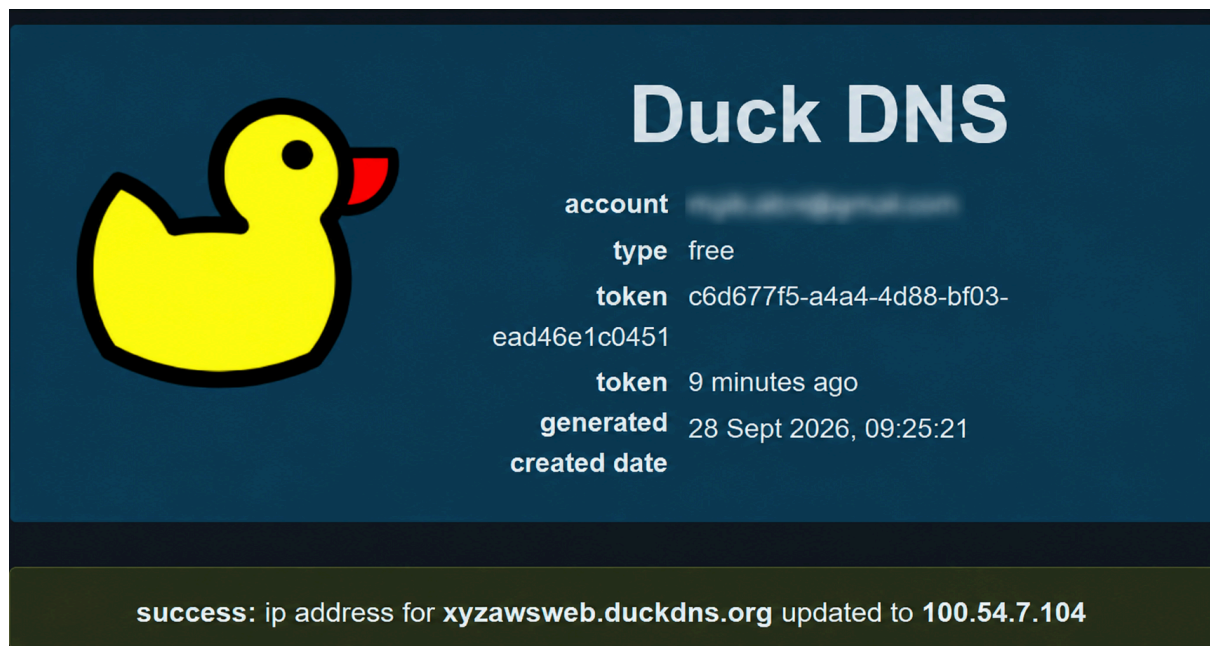
After installing the Apache webserver and editing, used the EC2 instance Public IP to check the webserver.



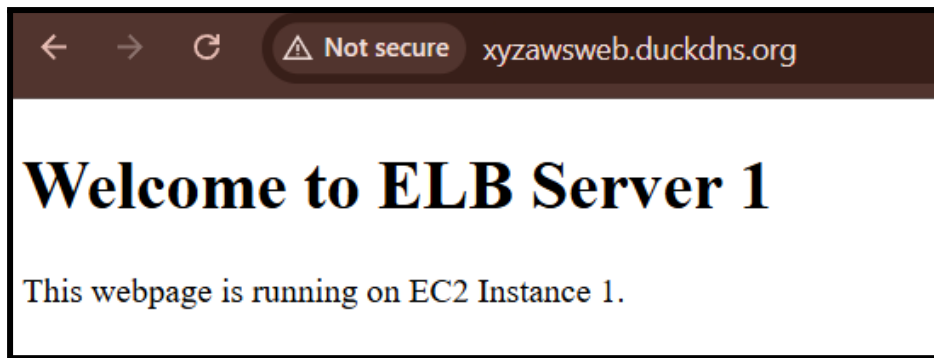
Created the free DuckDNS hostname

Created a free domain using DuckDNS : xyzawsserver.duckdns.org

Updated the ip address of the domain with public ip address of the EC2 instance.

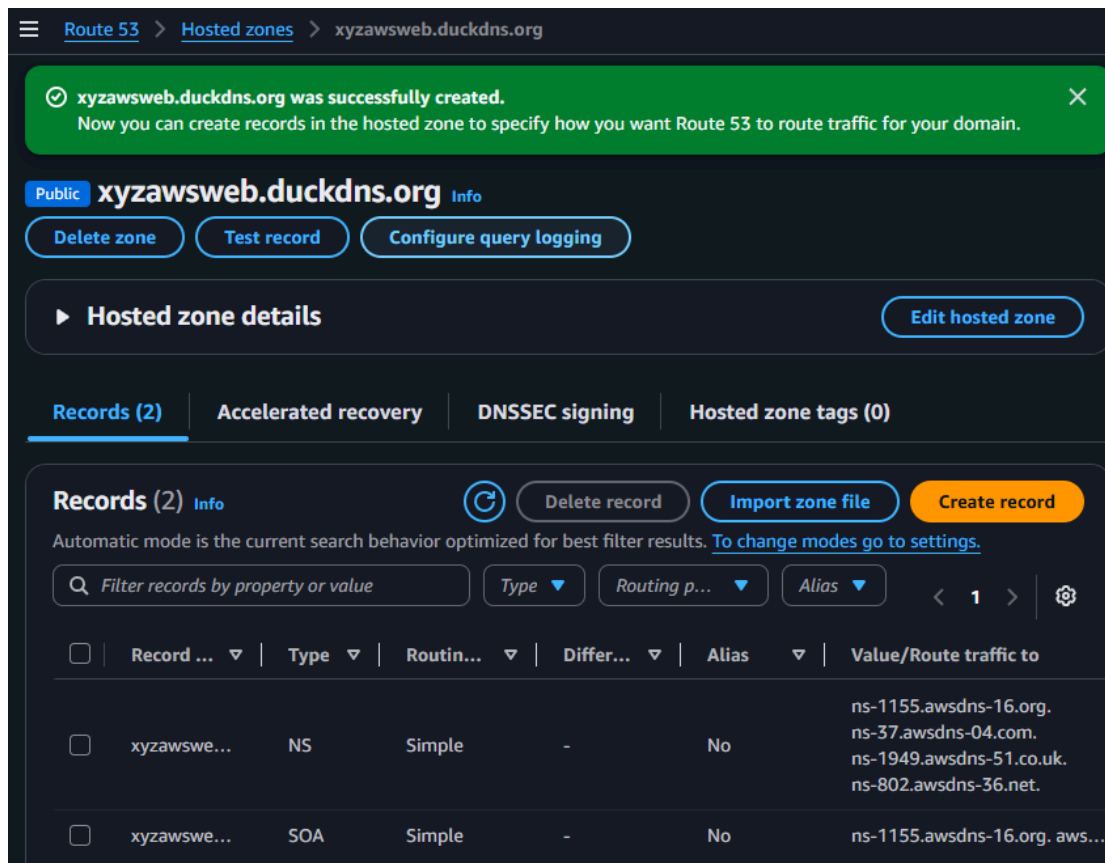


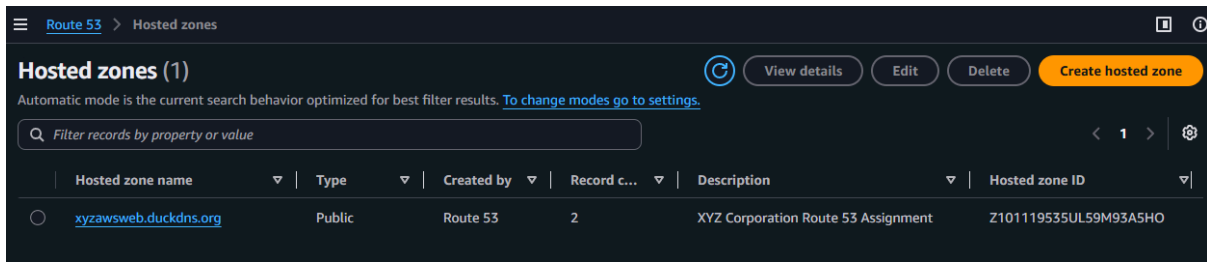
Tested the DuckDNS hostname



Task 1 — Used Route 53 hosted zone

AWS Console → Route 53 → Hosted zones → Create hosted zone





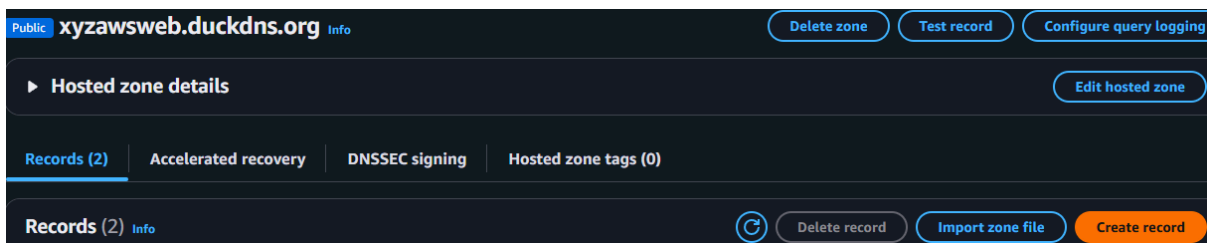
Task 2 — Routed traffic to EC2 Apache server using IP

Routed the traffic to an EC2 instance with an Apache web server running in it using its IP address.

Created the A record

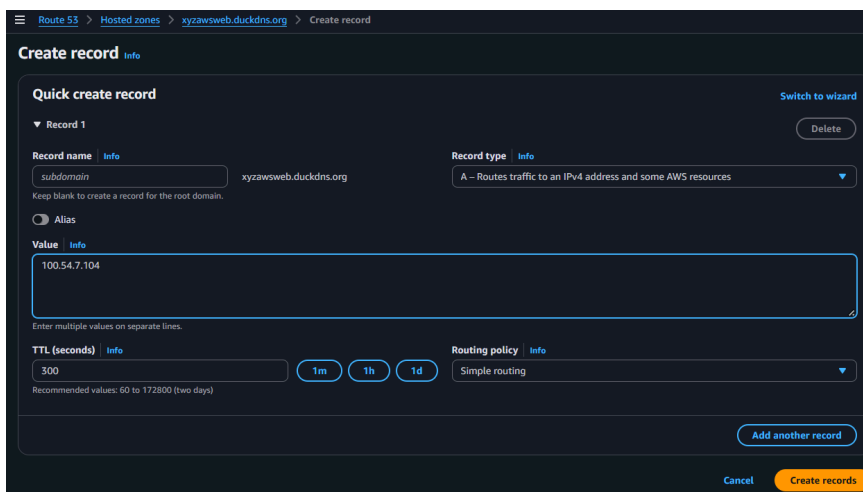
Clicked : **xyzawsweb.duckdns.org**

Then clicked **Create record**.



Configuration:

Record type : A
 Value : 100.54.7.104
 TTL : Leave default
 Routing policy : Simple routing



Then clicked **Create records**.

Route 53 > Hosted zones > xyzawswb.duckdns.org

Public xyzawswb.duckdns.org Info Delete zone Test record Configure query log

▶ Hosted zone details Edit hosted zone

Records (3) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (3) Info Delete record Import zone file Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Q Filter records by property or value Type Routing p... Alias < 1 >

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...
☐	xyzawswb.duckdns.org	A	Simple	-	No	100.54.7.104	300
☐	xyzawswb.duckdns.org	NS	Simple	-	No	ns-1155.awsdns-16.org. ns-37.awsdns-04.com. ns-1949.awsdns-51.co.uk. ns-802.awsdns-36.net.	172,800
☐	xyzawswb.duckdns.org	SOA	Simple	-	No	ns-1155.awsdns-16.org. aws...	900

A record successfully created.

Final browser verification

← → ↻ ⚠ Not secure xyzawswb.duckdns.org

Welcome to ELB Server 1

This webpage is running on EC2 Instance 1.